

Disassembler editor. In most cases, the user is allowed to reverse-trace the original program successfully from the original code using this editor. The editor supports loading of hexadecimal file format that is specific to Intel.

Assembler workspace. The workspace contains Address field, Label, Mnemonics, Hex-code, Mnemonic size, M-cycles and T-states. While it supports the static-timing diagram for all instructions, dynamic-timing diagram for step-by-step simulations are also supported. It also provides error checking facility.

Memory and I/O editors. The memory editor allows users to choose memory range and to modify data in a particular memory location. Users can either view the entire memory content or the one in the loaded memory location. They can store data directly into a specified memory location.

The I/O editor required for peripheral interfacing enables users to edit the content directly.

Interrupt editor. An interrupt is a mechanism by which an instruction suspends normal execution of the program and gets itself serviced. A non-maskable interrupt cannot be ignored by standard interrupt-masking techniques in the system, while a maskable interrupt can be disabled by writing some instruction. The simulator supports a non-maskable interrupt in 8085 (TRAP), maskable vectored interrupts (RST 7.5, RST 6.5 and RST 5.5) and an externally-serviced

Printing contents

The software allows the user to print contents of the assembler and the workspace. The equivalent binary value corresponding to the values of each register in the register bank can be printed. Values present in accumulator, registers B, C, D, E, H and L and memory (M) can all be retrieved.

Users can also access contents in Flag Register, Stack Pointer (SP), Memory Pointer (HL), Program Status Word (PSW), Program Counter (PC), Clock Cycle Counter, Instruction Counter and special blocks for monitoring Flag Register and usage of SIM and RIM instructions.

interrupt (INTR).

The interrupt editor allows triggering of these interrupts by pressing appropriate columns in the interrupt table.

Debugger. The debugger allows users to have a step-by-step debugging of programs by traversing a program both forwards and backwards. They can put break points and continue the program from break points.

Simulator. The simulator allows three levels of simulations. Users can have a step-by-step execution where they can halt between lines and simulate the code step by step. In the normal speed of execution, full execution takes place reflecting the intermittent states periodically. The ultimate execution reflects final state directly.

Supporting features

Crash recovery. Even if there is a sudden shutdown or crash of the application, the user can easily recover the program in this simulator.

Helper. Even if the user has very limited knowledge about user code, Code Wizard tool helps in programming 8085. The software is preloaded with a lot of sample programs

to aid new users. The user manual further aids these.

Tools. The simulator has Insert DELAY Subroutine tool that allows users to generate a user-defined delay with the help of registers at a particular operating frequency.

Memory values can be set at corresponding vector interrupt addresses with the help of Interrupt Service Subroutine tool.

Number Conversion Tool facilitates inter-conversion between hexadecimal, decimal and binary number formats.

Enhanced laboratory experience

The software is designed mainly for users who do not have sufficient hardware to learn microprocessor programming. It uses the same simulation engine at the back end and the user gets the same lab experience while coding in the simulator. Moreover, there are a lot of inbuilt sample programs to aid self-learning. If you are new to microprocessor programming and really interested in coding, give it a try. **EFY**

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